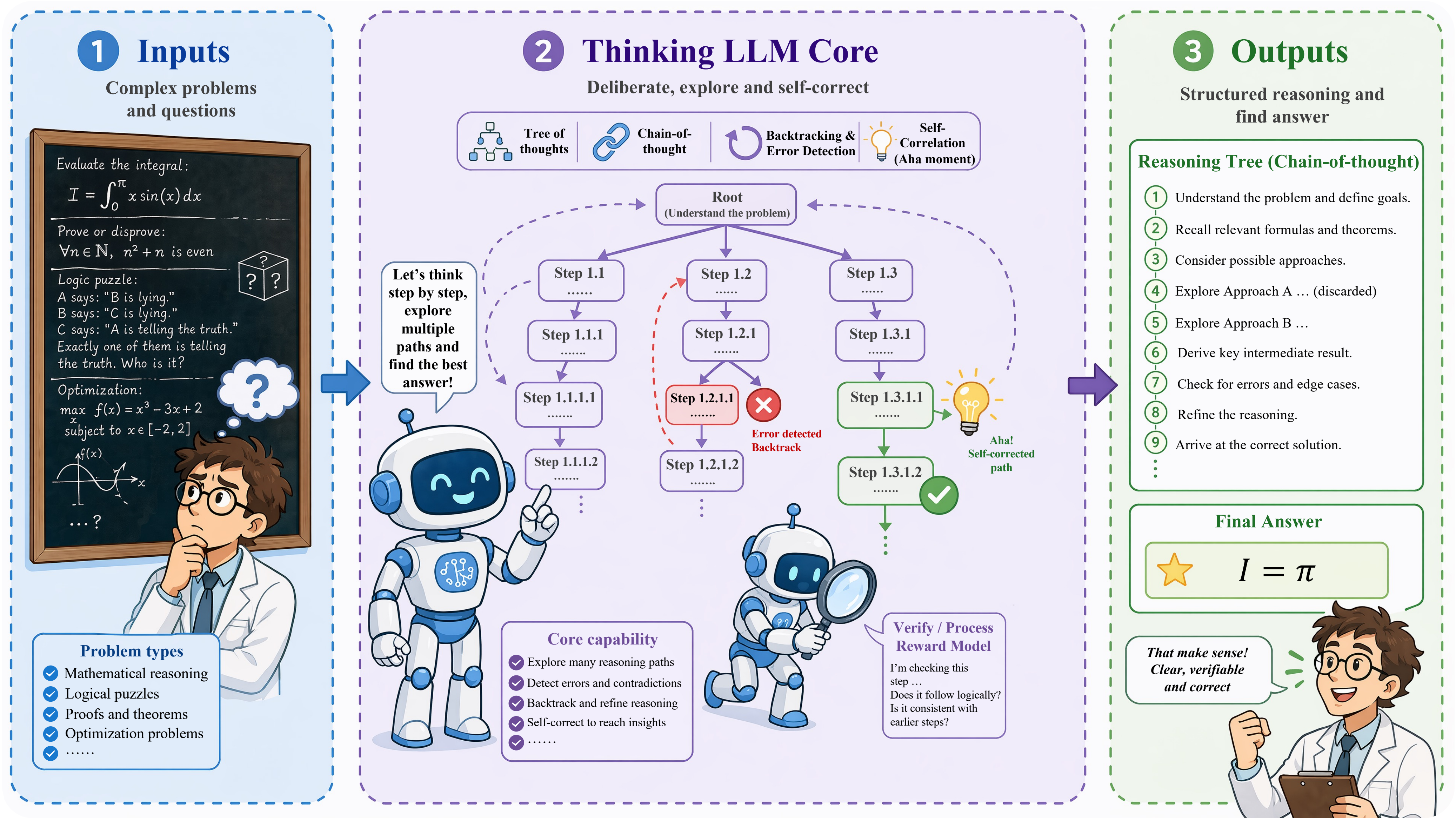
1 Inputs
Complex problems and questions
Evaluate the integral:
I = ∫₀^π x sin(x) dx
Prove or disprove:
∀n ∈ N, n² + n is even
Logic puzzle:
A says: "B is lying."
B says: "C is lying."
C says: "A is telling the truth."
Exactly one of them is telling the truth. Who is it?
Optimization:
max f(x) = x³ − 3x + 2
subject to x ∈ [−2, 2]
f(x)
x
… ?
Problem types
Mathematical reasoning
Logical puzzles
Proofs and theorems
Optimization problems
……
2 Thinking LLM Core
Deliberate, explore and self-correct
Tree of thoughts
Chain-of-thought
Backtracking & Error Detection
Self-Correlation (Aha moment)
Root (Understand the problem)
Let's think step by step, explore multiple paths and find the best answer!
Step 1.1
Step 1.2
Step 1.3
Step 1.1.1
Step 1.2.1
Step 1.3.1
Step 1.1.1.1
Step 1.2.1.1
Step 1.3.1.1
Error detected Backtrack
Aha! Self-corrected path
Step 1.1.1.2
Step 1.2.1.2
Step 1.3.1.2
Core capability
Explore many reasoning paths
Detect errors and contradictions
Backtrack and refine reasoning
Self-correct to reach insights
……
Verify / Process Reward Model
I'm checking this step …
Does it follow logically?
Is it consistent with earlier steps?
3 Outputs
Structured reasoning and find answer
Reasoning Tree (Chain-of-thought)
1 Understand the problem and define goals.
2 Recall relevant formulas and theorems.
3 Consider possible approaches.
4 Explore Approach A … (discarded)
5 Explore Approach B …
6 Derive key intermediate result.
7 Check for errors and edge cases.
8 Refine the reasoning.
9 Arrive at the correct solution.
Final Answer
I = π
That make sense! Clear, verifiable and correct